2. GLASS PLATE (10 points) — *Rasmus Kisel and Mihkel Heidelberg.* Light from a laser with a wavelength of λ and power of P falls perpendicularly on to a thin glass plate of thickness $a = (100.25\lambda)/n$ where n is the index of refraction of glass. The reflection coefficient of the glass surfaces is r, which can be interpreted as the probability for a photon to be reflected from a single glass surface.

i) *(2 points)* Calculate the force F_a applied to the glass by the laser light when the incident side of the glass is painted totally black.

ii) *(3 points)* Calculate the force F_b applied to the glass by the laser light when the opposite side of the glass is painted totally black.

iii) *(5 points)* Calculate the force F_c applied to the glass by the laser light when neither of the sides is painted.